



National  
Qualifications  
2024

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## **2024 Mathematics**

### **Higher - Paper 1**

## **Question Paper Finalised Marking Instructions**

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## General marking principles for Higher Mathematics

*Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.*

*For each question, the marking instructions are generally in two sections:*

- *generic scheme – this indicates why each mark is awarded*
- *illustrative scheme – this covers methods which are commonly seen throughout the marking*

*In general, you should use the illustrative scheme. Only use the generic scheme where a candidate has used a method not covered in the illustrative scheme.*

- Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.
- If you are uncertain how to assess a specific candidate response because it is not covered by the general marking principles or the detailed marking instructions, you must seek guidance from your team leader.
- One mark is available for each •. There are no half marks.
- If a candidate's response contains an error, all working subsequent to this error must still be marked. Only award marks if the level of difficulty in their working is similar to the level of difficulty in the illustrative scheme.
- Only award full marks where the solution contains appropriate working. A correct answer with no working receives no mark, unless specifically mentioned in the marking instructions.
- Candidates may use any mathematically correct method to answer questions, except in cases where a particular method is specified or excluded.
- If an error is trivial, casual or insignificant, for example  $6 \times 6 = 12$ , candidates lose the opportunity to gain a mark, except for instances such as the second example in point (h) below.
- If a candidate makes a transcription error (question paper to script or within script), they lose the opportunity to gain the next process mark, for example

This is a transcription error and so the mark is not awarded.

This is no longer a solution of a quadratic equation, so the mark is not awarded.

$$x^2 + 5x + 7 = 9x + 4$$

$$x - 4x + 3 = 0$$

$$x = 1$$

The following example is an exception to the above

This error is not treated as a transcription error, as the candidate deals with the intended quadratic equation. The candidate has been given the benefit of the doubt and all marks awarded.

$$x^2 + 5x + 7 = 9x + 4$$

$$x - 4x + 3 = 0$$

$$(x - 3)(x - 1) = 0$$

$$x = 1 \text{ or } 3$$

(i) **Horizontal/vertical marking**

If a question results in two pairs of solutions, apply the following technique, but only if indicated in the detailed marking instructions for the question.

Example:

$$\begin{array}{ccc} & \bullet^5 & \bullet^6 \\ \bullet^5 & x = 2 & x = -4 \\ \bullet^6 & y = 5 & y = -7 \end{array}$$

Horizontal:  $\bullet^5 x = 2$  and  $x = -4$     Vertical:  $\bullet^5 x = 2$  and  $y = 5$   
 $\bullet^6 y = 5$  and  $y = -7$                        $\bullet^6 x = -4$  and  $y = -7$

You must choose whichever method benefits the candidate, **not** a combination of both.

(j) In final answers, candidates should simplify numerical values as far as possible unless specifically mentioned in the detailed marking instruction. For example

$$\begin{array}{ll} \frac{15}{12} \text{ must be simplified to } \frac{5}{4} \text{ or } 1\frac{1}{4} & \frac{43}{1} \text{ must be simplified to } 43 \\ \frac{15}{0.3} \text{ must be simplified to } 50 & \frac{4\cancel{5}}{3} \text{ must be simplified to } \frac{4}{15} \\ \sqrt{64} \text{ must be simplified to } 8^* & \end{array}$$

\*The square root of perfect squares up to and including 144 must be known.

- (k) Commonly Observed Responses (CORs) are shown in the marking instructions to help mark common and/or non-routine solutions. CORs may also be used as a guide when marking similar non-routine candidate responses.
- (l) Do not penalise candidates for any of the following, unless specifically mentioned in the detailed marking instructions:
- working subsequent to a correct answer
  - correct working in the wrong part of a question
  - legitimate variations in numerical answers/algebraic expressions, for example angles in degrees rounded to nearest degree
  - omission of units
  - bad form (bad form only becomes bad form if subsequent working is correct), for example  
 $(x^3 + 2x^2 + 3x + 2)(2x + 1)$  written as  
 $(x^3 + 2x^2 + 3x + 2) \times 2x + 1$   
 $= 2x^4 + 5x^3 + 8x^2 + 7x + 2$  gains full credit
  - repeated error within a question, but not between questions or papers
- (m) In any 'Show that...' question, where candidates have to arrive at a required result, the last mark is not awarded as a follow-through from a previous error, unless specified in the detailed marking instructions.

- (n) You must check all working carefully, even where a fundamental misunderstanding is apparent early in a candidate's response. You may still be able to award marks later in the question so you must refer continually to the marking instructions. The appearance of the correct answer does not necessarily indicate that you can award all the available marks to a candidate.
- (o) You should mark legible scored-out working that has not been replaced. However, if the scored-out working has been replaced, you must only mark the replacement working.
- (p) If candidates make multiple attempts using the same strategy and do not identify their final answer, mark all attempts and award the lowest mark. If candidates try different valid strategies, apply the above rule to attempts within each strategy and then award the highest mark.

For example:

|                                                                    |                                                                    |
|--------------------------------------------------------------------|--------------------------------------------------------------------|
| Strategy 1 attempt 1 is worth 3 marks.                             | Strategy 2 attempt 1 is worth 1 mark.                              |
| Strategy 1 attempt 2 is worth 4 marks.                             | Strategy 2 attempt 2 is worth 5 marks.                             |
| From the attempts using strategy 1, the resultant mark would be 3. | From the attempts using strategy 2, the resultant mark would be 1. |

In this case, award 3 marks.

# Marking Instructions for each question

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  | Generic scheme                                                                                                                                                                                            | Illustrative scheme                                                                                                                                                                                                                                                | Max mark |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> use <math>m = \tan \theta</math></li> <li>•<sup>2</sup> evaluate exact value</li> <li>•<sup>3</sup> determine equation</li> </ul>                    | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>m = \tan 30^\circ</math></li> <li>•<sup>2</sup> <math>\frac{1}{\sqrt{3}}</math></li> <li>•<sup>3</sup> eg <math>y = \frac{1}{\sqrt{3}}x + 4</math> or <math>\sqrt{3}y - 4\sqrt{3} = x</math></li> </ul> | 3        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |  |                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                    |          |
| <p>1. Do not award •<sup>1</sup> for <math>m = \tan^{-1} 30^\circ</math>. However •<sup>2</sup> and •<sup>3</sup> are still available.</p> <p>2. Do not penalise the omission of a degree symbol at •<sup>1</sup>.</p> <p>3. Where candidates make no reference to a trigonometric ratio, or use an incorrect trigonometric ratio, •<sup>1</sup> and •<sup>2</sup> are unavailable. See Candidate A.</p> <p>4. •<sup>3</sup> is only available as a consequence of attempting to use a tan ratio. See Candidate F.</p> <p>5. •<sup>3</sup> is not available for using a gradient of 30.</p> <p>6. At •<sup>3</sup> accept any rearrangement of a candidate's equation where constant terms have been simplified.</p> <p>7. Accept <math>y - 4 = \frac{1}{\sqrt{3}}(x)</math> but not <math>y - 4 = \frac{1}{\sqrt{3}}(x - 0)</math> for •<sup>3</sup>.</p> |  |  |                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                    |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |  |                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                    |          |
| <b>Candidate A - no trig ratio</b><br>$m = \frac{1}{\sqrt{3}}$ • <sup>1</sup> ^    • <sup>2</sup> ✓ <sub>2</sub><br>$y = \frac{1}{\sqrt{3}}x + 4$ • <sup>3</sup> ✓ <sub>1</sub>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |  | <b>Candidate B</b><br>$m = \tan \theta$ • <sup>1</sup> ^<br>$y = \frac{1}{\sqrt{3}}x + 4$ • <sup>2</sup> ✓    • <sup>3</sup> ✓                                                                            |                                                                                                                                                                                                                                                                    |          |
| <b>Candidate C</b><br>$m = \tan \theta$ • <sup>1</sup> ^<br>$y = \sqrt{3}x + 4$ • <sup>2</sup> x    • <sup>3</sup> x                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                    |          |
| <b>Candidate D</b><br>$m = \tan \theta = 30$ • <sup>1</sup> x<br>$m = \frac{1}{\sqrt{3}}$ • <sup>2</sup> ✓ <sub>1</sub><br>$y = \frac{1}{\sqrt{3}}x + 4$ • <sup>3</sup> ✓ <sub>1</sub>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  | <b>Candidate E - no reference to m</b><br>$\tan 30^\circ = \frac{1}{\sqrt{3}}$ • <sup>2</sup> ✓<br>$y - 4 = \frac{1}{\sqrt{3}}(x - 0)$ • <sup>1</sup> ✓<br>$y = \frac{1}{\sqrt{3}}x + 4$ • <sup>3</sup> ✓ |                                                                                                                                                                                                                                                                    |          |
| <b>Candidate F - not using tan</b><br>$m = \sin 30^\circ$ • <sup>1</sup> x<br>$m = \frac{1}{2}$ • <sup>2</sup> ✓ <sub>2</sub><br>$y = \frac{1}{2}x + 4$ • <sup>3</sup> ✓ <sub>2</sub>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |  |                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                    |          |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |      | Generic scheme                                                                     | Illustrative scheme                                                                                                                     | Max mark |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | (a) |      | • <sup>1</sup> calculate second term                                               | • <sup>1</sup> 16                                                                                                                       | 1        |
| Notes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |      |                                                                                    |                                                                                                                                         |          |
| 1. Candidates who use $u_0 = 20$ and then calculate $u_1 = 16$ gain • <sup>1</sup> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |      |                                                                                    |                                                                                                                                         |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |      |                                                                                    |                                                                                                                                         |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | (b) | (i)  | • <sup>2</sup> communicate condition for limit to exist                            | • <sup>2</sup> a limit exists as $-1 < \frac{1}{5} < 1$                                                                                 | 1        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     | (ii) | • <sup>3</sup> know how to calculate a limit<br><br>• <sup>4</sup> calculate limit | • <sup>3</sup> $\frac{12}{1 - \frac{1}{5}}$ or $L = \frac{1}{5}L + 12$<br><br>• <sup>4</sup> 15                                         | 2        |
| Notes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |      |                                                                                    |                                                                                                                                         |          |
| 2. For • <sup>2</sup> accept:<br>any of ' $-1 < \frac{1}{5} < 1$ ' or ' $\left \frac{1}{5}\right  < 1$ ' or ' $0 < \frac{1}{5} < 1$ ' with no further comment;<br>or statements such as:<br>' $\frac{1}{5}$ lies between $-1$ and $1$ ' or ' $\frac{1}{5}$ is a proper fraction'.<br>3. • <sup>2</sup> is not available for:<br>' $-1 \leq \frac{1}{5} \leq 1$ ' or ' $\frac{1}{5} < 1$ '<br>or statements such as:<br>'It is between $-1$ and $1$ .' or ' $\frac{1}{5}$ is a fraction'.<br>4. Candidates who state $-1 < a < 1$ can only gain • <sup>2</sup> if it is explicitly stated that $a = \frac{1}{5}$ .<br>5. Do not accept $L = \frac{b}{1-a}$ with no further working for • <sup>3</sup> .<br>6. • <sup>3</sup> and • <sup>4</sup> are not available to candidates who conjecture $L = 15$ following the calculation of further terms in the sequence.<br>7. For $L = 15$ with no working award 0/2.<br>8. • <sup>4</sup> is only available where • <sup>3</sup> has been awarded. |     |      |                                                                                    |                                                                                                                                         |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |      |                                                                                    |                                                                                                                                         |          |
| Candidate A<br>$a = \frac{1}{5}$<br>$-1 < a < 1$ so a limit exists                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |      | • <sup>2</sup> ✓                                                                   | Candidate B - no explicit reference to $a$<br>$u_{n+1} = au_n + b$<br>$u_{n+1} = \frac{1}{5}u_n + 12$<br>$-1 < a < 1$ so a limit exists |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |      |                                                                                    | • <sup>2</sup> ^                                                                                                                        |          |



| Question                                                                              |  |  | Generic scheme                                                                                        | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Max mark |
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| 4.                                                                                    |  |  | <b>Method 1</b><br><br>• <sup>1</sup> interpret ratio<br><br>• <sup>2</sup> find coordinates of R     | <b>Method 1</b><br><br>• <sup>1</sup> $\begin{pmatrix} 2 \\ 4 \\ -4 \end{pmatrix}, \begin{pmatrix} -2 \\ -4 \\ 4 \end{pmatrix}, \begin{pmatrix} 3 \\ 6 \\ -6 \end{pmatrix}$ or $\begin{pmatrix} -3 \\ -6 \\ 6 \end{pmatrix}$<br><br>• <sup>2</sup> $(-4, 5, -2)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2        |
|                                                                                       |  |  | <b>Method 2</b><br><br>• <sup>1</sup> interpret ratio<br><br>• <sup>2</sup> find coordinates of R     | <b>Method 2</b><br><br>• <sup>1</sup> eg $\overline{PR} = \frac{2}{5}\overline{PQ}, \overline{QR} = \frac{3}{5}\overline{QP}$ or<br>$\overline{PR} = \frac{2}{3}\overline{RQ}$<br><br>• <sup>2</sup> $(-4, 5, -2)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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|                                                                                       |  |  | <b>Method 3</b><br><br>• <sup>1</sup> use section formula<br><br>• <sup>2</sup> find coordinates of R | <b>Method 3</b><br><br>• <sup>1</sup> $\frac{1}{5}(3\mathbf{p} + 2\mathbf{q})$<br><br>• <sup>2</sup> $(-4, 5, -2)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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| <b>Notes:</b>                                                                         |  |  |                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| 1. For $(-4, 5, -2)$ without working award 2/2.                                       |  |  |                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 2. For $\begin{pmatrix} -4 \\ 5 \\ -2 \end{pmatrix}$ without working award 1/2.       |  |  |                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| 3. For $(-3, 7, -4)$ (ratio of 3:2 with working) award 1/2. See Candidate A.          |  |  |                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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| 4. For $\begin{pmatrix} -3 \\ 7 \\ -4 \end{pmatrix}$ without working award 0/2.       |  |  |                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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| <b>Commonly Observed Responses:</b>                                                   |  |  |                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| <b>Candidate A</b><br>$\overline{PR} = \frac{3}{5}\overline{PQ}$<br>$R = (-3, 7, -4)$ |  |  | • <sup>1</sup> ✗<br><br>• <sup>2</sup> ✓ <sub>1</sub>                                                 | <b>Candidate B</b><br>$\frac{\overline{PR}}{\overline{RQ}} = \frac{2}{3}$<br>$3\overline{PR} = 2\overline{RQ}$<br>$3(\mathbf{r} - \mathbf{p}) = 2(\mathbf{q} - \mathbf{r})$<br>$5\mathbf{r} = 2\mathbf{q} + 3\mathbf{p}$<br>Leading to correct answer of<br>$R = (-4, 5, -2)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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✓<br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br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| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Generic scheme                                | Illustrative scheme                                                                                         | Max mark                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| 4. (continued)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                               |                                                                                                             |                                               |
| <p><b>Candidate C</b></p> $\overrightarrow{PQ} = \begin{pmatrix} 5 \\ 10 \\ -10 \end{pmatrix}$ $R = \begin{pmatrix} 2 \\ 4 \\ -4 \end{pmatrix}$ $R = \begin{pmatrix} -6 \\ 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 2 \\ 4 \\ -4 \end{pmatrix}$ $R = \begin{pmatrix} -4 \\ 5 \\ -2 \end{pmatrix}$ $R(-4, 5, -2)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>•<sup>1</sup> ✓</p> <p>•<sup>2</sup> ✓</p> | <p><b>Candidate D</b></p> $\overrightarrow{PR} = \begin{pmatrix} 2 \\ 4 \\ -4 \end{pmatrix}$ $R(-8, -3, 6)$ | <p>•<sup>1</sup> ✓</p> <p>•<sup>2</sup> ✗</p> |
| <p><b>Candidate E - stepping out using absolute values</b></p> <div style="text-align: center;"> <math>\begin{matrix} 2 &amp; : &amp; 3 \\ &amp; 5 &amp; \end{matrix}</math> </div> <div style="display: flex; justify-content: space-between; align-items: center;"> <span>-6</span> <span>-1</span> </div> <div style="text-align: center;"> <math>\begin{matrix} 2 &amp; \text{or} &amp; 3 \\ &amp; 10 &amp; \end{matrix}</math> </div> <div style="display: flex; justify-content: space-between; align-items: center;"> <span>1</span> <span>11</span> </div> <div style="text-align: center;"> <math>\begin{matrix} 4 &amp; \text{or} &amp; 6 \\ &amp; 10 &amp; \end{matrix}</math> </div> <div style="display: flex; justify-content: space-between; align-items: center;"> <span>2</span> <span>-8</span> </div> $R(-4, 5, -2)$ | <p>•<sup>1</sup> ✓</p> <p>•<sup>2</sup> ✓</p> |                                                                                                             |                                               |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  | Generic scheme                                                                                                                                                                                                                                                      | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                    | Max mark |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 5.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |  | <b>Method 1</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> equate composite function to <math>x</math></li> <li>•<sup>2</sup> write <math>h(h^{-1}(x))</math> in terms of <math>h^{-1}(x)</math></li> <li>•<sup>3</sup> state inverse function</li> </ul> | <b>Method 1</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>h(h^{-1}(x)) = x</math></li> <li>•<sup>2</sup> <math>2(h^{-1}(x))^3 - 7 = x</math></li> <li>•<sup>3</sup> <math>h^{-1}(x) = \sqrt[3]{\frac{x+7}{2}}</math></li> </ul>                                                                                                                       | 3        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |  | <b>Method 2</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> write as <math>y = h(x)</math> and start to rearrange</li> <li>•<sup>2</sup> express <math>x</math> in terms of <math>y</math></li> <li>•<sup>3</sup> state inverse function</li> </ul>        | <b>Method 2</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>y = h(x) \Rightarrow x = h^{-1}(y)</math><br/><math>y + 7 = 2x^3</math></li> <li>•<sup>2</sup> <math>x = \sqrt[3]{\frac{y+7}{2}}</math></li> <li>•<sup>3</sup> <math>h^{-1}(y) = \sqrt[3]{\frac{y+7}{2}}</math><br/><math>\Rightarrow h^{-1}(x) = \sqrt[3]{\frac{x+7}{2}}</math></li> </ul> |          |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |  |                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                        |          |
| 1. In method 1, accept $2(h^{-1}(x))^3 - 7 = x$ for • <sup>1</sup> and • <sup>2</sup> .<br>2. In method 2, accept ' $y + 7 = 2x^3$ ', without reference to $y = h(x) \Rightarrow x = h^{-1}(y)$ at • <sup>1</sup> .<br>3. In method 2, accept $h^{-1}(x) = \sqrt[3]{\frac{x+7}{2}}$ without reference to $h^{-1}(y)$ at • <sup>3</sup> .<br>4. In method 2, beware of candidates with working where each line is not mathematically equivalent. See candidates A and B for example.<br>5. At • <sup>3</sup> stage, accept $h^{-1}$ written in terms of any dummy variable.<br>For example $h^{-1}(y) = \sqrt[3]{\frac{y+7}{2}}$ .<br>6. $y = \sqrt[3]{\frac{x+7}{2}}$ does not gain • <sup>3</sup> .<br>7. $h^{-1}(x) = \sqrt[3]{\frac{x+7}{2}}$ with no working gains 3/3. |  |  |                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                        |          |

| Question                                                                                                                                                                                                                        | Generic scheme                                                             | Illustrative scheme                                                                                                                                   | Max mark                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>5. (continued)</b>                                                                                                                                                                                                           |                                                                            |                                                                                                                                                       |                                                                            |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                             |                                                                            |                                                                                                                                                       |                                                                            |
| <b>Candidate A</b><br>$h(x) = 2x^3 - 7$<br>$y = 2x^3 - 7$<br>$x = \sqrt[3]{\frac{y+7}{2}}$<br>$y = \sqrt[3]{\frac{x+7}{2}}$<br>$h^{-1}(x) = \sqrt[3]{\frac{x+7}{2}}$                                                            | $\bullet^1 \checkmark \bullet^2 \checkmark$<br>$\bullet^3 \times$          | <b>Candidate B</b><br>$h(x) = 2x^3 - 7$<br>$y = 2x^3 - 7$<br>$x = 2y^3 - 7$<br>$y = \sqrt[3]{\frac{x+7}{2}}$<br>$h^{-1}(x) = \sqrt[3]{\frac{x+7}{2}}$ | $\bullet^1 \times$<br>$\bullet^2 \checkmark_1$<br>$\bullet^3 \checkmark_1$ |
| <b>Candidate C</b><br>$x = 2h(x)^3 - 7$<br>$h(x) = \sqrt[3]{\frac{x+7}{2}}$<br>$h^{-1}(x) = \sqrt[3]{\frac{x+7}{2}}$                                                                                                            | $\bullet^1 \times$<br>$\bullet^2 \checkmark_1$<br>$\bullet^3 \checkmark_1$ | <b>Candidate D - Method 1</b><br>$h(h^{-1}(x)) = 2(h^{-1}(x))^3 - 7$<br>$x = 2(h^{-1}(x))^3 - 7$<br>$h^{-1}(x) = \sqrt[3]{\frac{x+7}{2}}$             | $\bullet^2 \checkmark$<br>$\bullet^1 \checkmark$<br>$\bullet^3 \checkmark$ |
| <b>Candidate E</b><br>$x \rightarrow x^3 \rightarrow 2x^3 \rightarrow 2x^3 - 7 = h(x)$<br>$\times 2 \rightarrow -7$<br>$\therefore +7 \rightarrow \div 2$<br>$\sqrt[3]{\frac{x+7}{2}}$<br>$h^{-1}(x) = \sqrt[3]{\frac{x+7}{2}}$ | $\bullet^1 \checkmark$<br>$\bullet^2 \checkmark$<br>$\bullet^3 \checkmark$ | <b>Candidate F - BEWARE of incorrect notation</b><br>$h'(x) =$                                                                                        | $\bullet^3 \times$                                                         |

| Question |     |      | Generic scheme                                                                                                                                                                                                       | Illustrative scheme                                                                                                                                                                                                                                                                   | Max mark |
|----------|-----|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 6.       | (a) | (i)  | <ul style="list-style-type: none"> <li>•<sup>1</sup> find value of <math>\cos p</math></li> <li>•<sup>2</sup> substitute into the formula for <math>\sin 2p</math></li> <li>•<sup>3</sup> simplify answer</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\cos p = \frac{2}{\sqrt{5}}</math> stated or implied by •<sup>2</sup></li> <li>•<sup>2</sup> <math>2 \times \frac{1}{\sqrt{5}} \times \frac{2}{\sqrt{5}}</math></li> <li>•<sup>3</sup> <math>\frac{4}{5}</math></li> </ul> | 3        |
|          |     | (ii) | <ul style="list-style-type: none"> <li>•<sup>4</sup> evaluate <math>\cos 2p</math></li> </ul>                                                                                                                        | <ul style="list-style-type: none"> <li>•<sup>4</sup> <math>\frac{3}{5}</math></li> </ul>                                                                                                                                                                                              | 1        |

#### Notes:

- Evidence for •<sup>1</sup> may appear in (a)(ii).
- Where a candidate substitutes an incorrect value for  $\cos p$  at •<sup>2</sup>, •<sup>2</sup> may be awarded if the candidate has previously stated this incorrect value or it can be implied by a diagram or Pythagoras calculation. See Candidates A and B.
- Where a candidate explicitly states a value for  $\cos p$ , subsequent working must follow from that value for •<sup>2</sup> to be awarded.
- <sup>3</sup> is only available as a consequence of substituting into a valid formula at •<sup>2</sup>.
- Do not penalise trigonometric ratios which are less than -1 or greater than 1 throughout this question.

#### Commonly Observed Responses:

|                                                                                                                                             |  |  |                                                                                                                                 |  |  |
|---------------------------------------------------------------------------------------------------------------------------------------------|--|--|---------------------------------------------------------------------------------------------------------------------------------|--|--|
| Candidate A - incorrect use of Pythagoras                                                                                                   |  |  | Candidate B - no evidence of Pythagoras                                                                                         |  |  |
| $\sqrt{\sqrt{5}^2 + 1^2} = \sqrt{6}$<br>$2 \times \frac{1}{\sqrt{5}} \times \frac{\sqrt{6}}{\sqrt{5}}$<br>$\frac{2\sqrt{6}}{5}$             |  |  | $2 \times \frac{1}{\sqrt{5}} \times \frac{\sqrt{6}}{\sqrt{5}}$<br>$\frac{2\sqrt{6}}{5}$                                         |  |  |
| <ul style="list-style-type: none"> <li>•<sup>1</sup> ✗</li> <li>•<sup>2</sup> ✓<sub>1</sub></li> <li>•<sup>3</sup> ✓<sub>1</sub></li> </ul> |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> ^</li> <li>•<sup>2</sup> ✗</li> <li>•<sup>3</sup> ✓<sub>1</sub></li> </ul> |  |  |
| Candidate C                                                                                                                                 |  |  |                                                                                                                                 |  |  |
| $2 \times \sin \frac{1}{\sqrt{5}} \times \cos \frac{2}{\sqrt{5}}$<br>$\frac{4}{5}$                                                          |  |  |                                                                                                                                 |  |  |
| <ul style="list-style-type: none"> <li>•<sup>1</sup> ✓</li> <li>•<sup>2</sup> ✗</li> <li>•<sup>3</sup> ✗</li> </ul>                         |  |  |                                                                                                                                 |  |  |

|  |     |  |                                                                                               |                                                                                            |   |
|--|-----|--|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|---|
|  | (b) |  | <ul style="list-style-type: none"> <li>•<sup>5</sup> evaluate <math>\sin 4p</math></li> </ul> | <ul style="list-style-type: none"> <li>•<sup>5</sup> <math>\frac{24}{25}</math></li> </ul> | 1 |
|--|-----|--|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|---|

#### Notes:

- <sup>5</sup> is only available for an answer expressed as a single fraction.

#### Commonly Observed Responses:

|  |  |  |  |  |  |
|--|--|--|--|--|--|
|  |  |  |  |  |  |
|--|--|--|--|--|--|

| Question |  |  | Generic scheme                                                                                                                                                                                                                                                                                                                                            | Illustrative scheme                                                                                                                                                                                                                                                                          | Max mark |
|----------|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 7.       |  |  | <b>Method 1</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> substitute for <math>y</math></li> <li>•<sup>2</sup> write in standard quadratic form</li> <li>•<sup>3</sup> determine <math>x</math>-coordinate</li> <li>•<sup>4</sup> determine <math>y</math>-coordinate</li> </ul>                                                               | <b>Method 1</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>x^2 + (2x)^2 - 14x - 8(2x) + 45 = 0</math></li> <li>•<sup>2</sup> <math>5x^2 - 30x + 45 = 0</math></li> <li>•<sup>3</sup> 3</li> <li>•<sup>4</sup> 6</li> </ul>                                                   | 4        |
|          |  |  | <b>Method 2</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> substitute for <math>x</math></li> <li>•<sup>2</sup> write in standard quadratic form</li> <li>•<sup>3</sup> determine <math>y</math>-coordinate</li> <li>•<sup>4</sup> determine <math>x</math>-coordinate</li> </ul>                                                               | <b>Method 2</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\left(\frac{y}{2}\right)^2 + y^2 - 14\left(\frac{y}{2}\right) - 8y + 45 = 0</math></li> <li>•<sup>2</sup> <math>\frac{5}{4}y^2 - 15y + 45 = 0</math></li> <li>•<sup>3</sup> 6</li> <li>•<sup>4</sup> 3</li> </ul> |          |
|          |  |  | <b>Method 3</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> use centre and perpendicular gradient to determine equation of radius through point of contact</li> <li>•<sup>2</sup> substitute for <math>y</math></li> <li>•<sup>3</sup> determine <math>x</math>-coordinate</li> <li>•<sup>4</sup> determine <math>y</math>-coordinate</li> </ul> | <b>Method 3</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>x + 2y = 15</math></li> <li>•<sup>2</sup> <math>x + 2(2x) = 15</math></li> <li>•<sup>3</sup> 3</li> <li>•<sup>4</sup> 6</li> </ul>                                                                                |          |

**Notes:**

1. In Methods 1 and 2, treat an absence of brackets at the •<sup>1</sup> stage as bad form only if corrected on the next line of working.
2. In Methods 1 and 2, •<sup>1</sup> is only available if the '=0' appears by the •<sup>2</sup> stage.
3. In Methods 1 and 2, if a candidate arrives at an equation which is not a quadratic •<sup>3</sup> and •<sup>4</sup> are unavailable.
4. Where the quadratic obtained at •<sup>2</sup> in Methods 1 and 2, does not have repeated roots •<sup>3</sup> and •<sup>4</sup> are not available.
5. In Method 3 accept  $y - 4 = -\frac{1}{2}(x - 7)$ ,  $-\frac{1}{2} = \frac{4 - y}{7 - x}$  or equivalent for •<sup>1</sup>.
6. In Method 3 •<sup>2</sup>, •<sup>3</sup> and •<sup>4</sup> are unavailable to candidates who find the equation of any other line.
7. For (3,6) without working, award 0/4.
8. For answer of (3,6) verified in both equations, or (3,6) generated by the linear equation and verified in the equation of the circle, award 4/4.

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Generic scheme | Illustrative scheme | Max mark |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------|----------|
| <b>7. (continued)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                |                     |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |                     |          |
| <p><b>Candidate A - substitution into the equation of the circle</b></p> <p>∴</p> <p><math>x = 3</math> <span style="float: right;">•<sup>3</sup> ✓</span></p> <p><math>(3)^2 + y^2 - 14(3) - 8y + 45 = 0</math></p> <p><math>y^2 - 8y + 12 = 0</math></p> <p><math>(y - 2)(y - 6) = 0</math></p> <p><math>y = 6</math> <span style="float: right;">•<sup>4</sup> ✓</span></p> <p>no need to explicitly consider <math>y = 2</math></p> <p><b>However,</b></p> <p><math>(3, 6)</math> and <math>(3, 2)</math> <span style="float: right;">•<sup>4</sup> ✗</span></p> |                |                     |          |

| Question |  |  | Generic scheme                                                                                                                                                                                                                              | Illustrative scheme                                                                                                                                                                                                                                                                           | Max mark |
|----------|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 8.       |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> use discriminant</li> <li>•<sup>2</sup> apply condition</li> <li>•<sup>3</sup> identify roots of quadratic expression</li> <li>•<sup>4</sup> state range with justification</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>(m-4)^2 - 4(1)(2m-3)</math></li> <li>•<sup>2</sup> <math>(m-4)^2 - 4(1)(2m-3) &lt; 0</math></li> <li>•<sup>3</sup> 2, 14</li> <li>•<sup>4</sup> <math>2 &lt; m &lt; 14</math> with eg labelled sketch or table of signs</li> </ul> | 4        |

#### Notes:

- At •<sup>1</sup>, treat the inconsistent use of brackets: For example  $m - 4^2 - 4(1)(2m-3)$  or  $(m-4)^2 - 4 \times 1 \times 2m - 3$  as bad form only if the candidate deals with the unbracketed terms correctly in the next line of working.
- Where candidates express  $a$ ,  $b$  and  $c$  in terms of  $m$ , and then state  $b^2 - 4ac < 0$ , award •<sup>2</sup>.
- If candidates have the condition 'discriminant  $> 0$ ', 'discriminant  $\leq 0$ ' or 'discriminant  $\geq 0$ ', then •<sup>2</sup> is lost but •<sup>3</sup> and •<sup>4</sup> are available.
- Ignore the appearance of  $b^2 - 4ac = 0$  where the correct condition has subsequently been applied.
- If candidates only work with the condition 'discriminant  $= 0$ ', then •<sup>2</sup> and •<sup>4</sup> are unavailable.
- Accept the appearance of 2 and 14 within inequalities for •<sup>3</sup>.
- At •<sup>4</sup> accept " $m > 2$  and  $m < 14$ " or " $m > 2, m < 14$ " together with the required justification.
- For the appearance of  $x$  in any expression of the discriminant, no further marks are available.

#### Commonly Observed Responses:

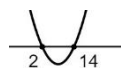
##### Candidate A - no expressions for $a$ , $b$ and $c$

No real roots  $b^2 - 4ac < 0$

$$m^2 - 16m + 28 = 0$$

$$m = 2, m = 14$$

$$2 < m < 14$$



•<sup>1</sup> ✓  
•<sup>3</sup> ✓  
•<sup>2</sup> ✓ •<sup>4</sup> ✓

In this case •<sup>2</sup> is only available  
where •<sup>4</sup> is awarded

##### Candidate B

$$(m-4)^2 - 4(1)(2m-3)$$

$$m^2 - 16m + 28 = 0$$

$$m = 2, m = 14$$

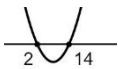
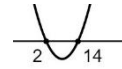
$$b^2 - 4ac < 0$$

$$2 < m < 14$$



•<sup>1</sup> ✓  
•<sup>3</sup> ✓  
•<sup>2</sup> ✓ •<sup>4</sup> ✓

In this case •<sup>2</sup> is only available  
where •<sup>4</sup> is awarded

| Question                                                                                                                                                                                                                                                                                                 | Generic scheme                                                                                                                                                                                                                                                               | Illustrative scheme | Max mark |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------|
| 8. (continued)                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                              |                     |          |
| <b>Candidate C</b><br>$(m-4)^2 - 4(1)(2m-3)$ • <sup>1</sup> ✓<br>$b^2 - 4ac = 0$<br>$m^2 - 16m + 28 = 0$<br>$m = 2, m = 14$ • <sup>3</sup> ✓<br>$m^2 - 16m + 28 < 0$ • <sup>2</sup> ✓<br>$2 < m < 14$ • <sup>4</sup> ✓  | <b>Candidate D</b><br>$(m-4)^2 - 4(1)(2m-3)$ • <sup>1</sup> ✓<br>$m^2 - 16m + 28 = 0$ • <sup>2</sup> ✗<br>$m = 2, m = 14$ • <sup>3</sup> ✓<br>$2 < m < 14$ • <sup>4</sup> ✓ <sub>2</sub>  |                     |          |
| <b>Candidate E - not solving a quadratic</b><br>$m - 4^2 - 4(1)(2m-3) < 0$ • <sup>1</sup> ✗ • <sup>2</sup> ✓ • <sup>3</sup> ✗<br>$-7m - 4 < 0$<br>$m > -\frac{4}{7}$ • <sup>4</sup> ✓ <sub>2</sub>                                                                                                       |                                                                                                                                                                                                                                                                              |                     |          |



| Question |  |  | Generic scheme                                                                                                                                                                                                                      | Illustrative scheme                                                                                                                                                                                                                    | Max mark |
|----------|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 9.       |  |  | <b>Method 1</b><br>• <sup>1</sup> apply $\log_a x + \log_a y = \log_a xy$<br><br>• <sup>2</sup> apply $m \log_a x = \log_a x^m$<br><br>• <sup>3</sup> apply $\log_a x - \log_a y = \log_a \frac{x}{y}$ and express in required form | <b>Method 1</b><br>• <sup>1</sup> $\log_a (5 \times 80) \dots$ stated or implied by • <sup>3</sup><br><br>• <sup>2</sup> $\dots - \log_a 10^2$ stated or implied by • <sup>3</sup><br><br>• <sup>3</sup> $\log_a 4$                    | 3        |
|          |  |  | <b>Method 2</b><br>• <sup>1</sup> apply $m \log_a x = \log_a x^m$<br><br>• <sup>2</sup> apply $\log_a x - \log_a y = \log_a \frac{x}{y}$<br><br>• <sup>3</sup> apply $\log_a x + \log_a y = \log_a xy$ and express in required form | <b>Method 2</b><br>• <sup>1</sup> $\dots - \log_a 10^2$ stated or implied by • <sup>3</sup><br><br>• <sup>2</sup> $\dots + \log_a \left( \frac{80}{10^2} \right)$ stated or implied by • <sup>3</sup><br><br>• <sup>3</sup> $\log_a 4$ |          |

#### Notes:

- Where an error at the •<sup>1</sup> or •<sup>2</sup> stage leads to a non-integer value for  $k$ , •<sup>3</sup> is still available.
- Each line of working must be equivalent to the line above within a valid strategy. See commonly observed responses.
- Where candidates apply the laws of logarithms in the incorrect order see Candidates A and B.
- Where candidates do not consider the '2', a maximum of 1/3 is available. See Candidate C.
- Do not penalise the omission of the base of the logarithm.
- Correct answer with no working, award 3/3.
- Where candidates form an invalid equation, •<sup>1</sup> and •<sup>2</sup> may only be awarded for working with  $\log_a 5 + \log_a 80 - 2 \log_a 10$  on one side of the equation; •<sup>3</sup> is not available.

#### Commonly Observed Responses:

##### Candidate A

$$\log_a 5 + 2 \log_a \left( \frac{80}{10} \right)$$

$$2 \log_a \left( \frac{5 \times 80}{10} \right)$$

$$\log_a (40)^2$$

$$\log_a 1600$$

**Award 1/3**

##### Candidate B

$$\log_a 400 - 2 \log_a 10$$

$$2 \log_a \left( \frac{400}{10} \right)$$

$$\log_a (40)^2$$

$$\log_a 1600$$

**Award 2/3**

##### Candidate C - ignoring the 2

$$\log_a 5 + \log_a 80 - 2 \log_a 10$$

$$\log_a 5 + \log_a \frac{80}{10}$$

$$\log_a 40$$

**Award 1/3**

| Question |     |  | Generic scheme                                                                                                                                      | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Max mark |
|----------|-----|--|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 10.      | (a) |  | <p>•<sup>1</sup> use 1 in synthetic division or in evaluation of quartic</p> <p>•<sup>2</sup> complete division/evaluation and interpret result</p> | <p>•<sup>1</sup></p> $\begin{array}{r rrrrr} 1 & 2 & 3 & -4 & -3 & 2 \\ & & 2 & & & \end{array}$ <p>or <math>2 \times (1)^4 + 3 \times (1)^3 - 4 \times (1)^2 - 3 \times (1) + 2</math></p> <p>•<sup>2</sup></p> $\begin{array}{r rrrrr} 1 & 2 & 3 & -4 & -3 & 2 \\ & & 2 & 5 & 1 & -2 \\ \hline & 2 & 5 & 1 & -2 & 0 \end{array}$ <p>Remainder = 0 <math>\therefore (x-1)</math> is a factor</p> <p>or</p> <p><math>f(1) = 0 \therefore (x-1)</math> is a factor</p> | 2        |

#### Notes:

- Communication at •<sup>2</sup> must be consistent with working at that stage i.e. a candidate's working must arrive legitimately at 0 before •<sup>2</sup> can be awarded.
- Accept any of the following for •<sup>2</sup>:
  - ' $f(1) = 0$  so  $(x-1)$  is a factor'
  - 'since remainder = 0, it is a factor'
  - the '0' from any method linked to the word 'factor' by 'so', 'hence',  $\therefore$ ,  $\rightarrow$ ,  $\Rightarrow$  etc.
- Do not accept any of the following for •<sup>2</sup>:
  - double underlining the '0' or boxing the '0' without comment
  - ' $x=1$  is a factor', '... is a root'
  - the word 'factor' only, with no link.

#### Commonly Observed Responses:

##### Candidate A - grid method

|      |         |        |  |  |
|------|---------|--------|--|--|
|      | $2x^3$  |        |  |  |
| $x$  | $2x^4$  | $5x^3$ |  |  |
| $-1$ | $-2x^3$ |        |  |  |

•<sup>1</sup> ✓

|      |         |         |       |       |
|------|---------|---------|-------|-------|
|      | $2x^3$  | $5x^2$  | $x$   | $-2$  |
| $x$  | $2x^4$  | $5x^3$  | $x^2$ | $-2x$ |
| $-1$ | $-2x^3$ | $-5x^2$ | $-x$  | $2$   |

'with no remainder'

$\therefore (x-1)$  is a factor •<sup>2</sup> ✓

##### Candidate B - grid method

|      |         |        |  |  |
|------|---------|--------|--|--|
|      | $2x^3$  |        |  |  |
| $x$  | $2x^4$  | $5x^3$ |  |  |
| $-1$ | $-2x^3$ |        |  |  |

•<sup>1</sup> ✓

|      |         |         |       |       |
|------|---------|---------|-------|-------|
|      | $2x^3$  | $5x^2$  | $x$   | $-2$  |
| $x$  | $2x^4$  | $5x^3$  | $x^2$ | $-2x$ |
| $-1$ | $-2x^3$ | $-5x^2$ | $-x$  | $2$   |

$\therefore (x-1)(2x^3 + 5x^2 + x - 2) = 2x^4 + 3x^3 - 4x^2 - 3x + 2$

$\therefore (x-1)$  is a factor •<sup>2</sup> ✓

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |  | Generic scheme                                                                                                                                                                           | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Max mark |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 10.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | (b) |  | <p>•<sup>3</sup> identify cubic and attempt to factorise</p> <p>•<sup>4</sup> find second factor</p> <p>•<sup>5</sup> identify quadratic</p> <p>•<sup>6</sup> complete factorisation</p> | <p>•<sup>3</sup> eg</p> $\begin{array}{r rrrr} -1 & 2 & 5 & 1 & -2 \\ & & -2 & -3 & \\ \hline & 2 & 3 & & \end{array}$ <p>or</p> $\begin{array}{r rrrr} -2 & 2 & 5 & 1 & -2 \\ & & -4 & -2 & \\ \hline & 2 & 1 & & \end{array}$ <p>•<sup>4</sup> eg</p> $\begin{array}{r rrrr} -1 & 2 & 5 & 1 & -2 \\ & & -2 & -3 & 2 \\ \hline & 2 & 3 & -2 & 0 \end{array}$ <p>leading to <math>(x+1)</math></p> <p>or</p> $\begin{array}{r rrrr} -2 & 2 & 5 & 1 & -2 \\ & & -4 & -2 & 2 \\ \hline & 2 & 1 & -1 & 0 \end{array}$ <p>leading to <math>(x+2)</math></p> <p>•<sup>5</sup> <math>2x^2 + 3x - 2</math> or <math>2x^2 + x - 1</math></p> <p>•<sup>6</sup> <math>(x-1)(x+1)(2x-1)(x+2)</math><br/>stated explicitly</p> | 4        |
| <b>Notes:</b> <p>4. Ignore the appearance of '<math>= 0</math>'.</p> <p>5. Candidates who arrive at <math>(x-1)(x+1)(2x^2 + 3x - 2)</math> or <math>(x-1)(x+2)(2x^2 + x - 1)</math> by using algebraic long division or by inspection, gain •<sup>3</sup>, •<sup>4</sup> and •<sup>5</sup>.</p> <p>6. Where a candidate only identifies additional factors from a quartic, only •<sup>4</sup> is available.</p> <p>7. •<sup>3</sup> and •<sup>4</sup> may be awarded for applications of synthetic division even when previous trials contain errors. •<sup>5</sup> and •<sup>6</sup> are available.</p> |     |  |                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |

| Question                                                                                                                                                                                                            | Generic scheme | Illustrative scheme                                                                                                                                                                                                 | Max mark |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 10. (b) (continued)                                                                                                                                                                                                 |                |                                                                                                                                                                                                                     |          |
| Commonly Observed Responses:                                                                                                                                                                                        |                |                                                                                                                                                                                                                     |          |
| Candidate C - grid method                                                                                                                                                                                           |                | Candidate D - grid method                                                                                                                                                                                           |          |
| (a)                                                                                                                                                                                                                 |                | (a)                                                                                                                                                                                                                 |          |
| $  \begin{array}{r rrrr}  & 2x^3 & 5x^2 & x & -2 \\  x & 2x^4 & 5x^3 & x^2 & -2x \\  -1 & -2x^3 & -5x^2 & -x & 2  \end{array}  $                                                                                    |                | $  \begin{array}{r rrrr}  & 2x^3 & 5x^2 & x & -2 \\  x & 2x^4 & 5x^3 & x^2 & -2x \\  -1 & -2x^3 & -5x^2 & -x & 2  \end{array}  $                                                                                    |          |
| (b)                                                                                                                                                                                                                 |                | (b)                                                                                                                                                                                                                 |          |
| $  \begin{array}{r rrrr}  & 2x^2 & \dots & \dots & \\  x & 2x^3 & \dots & \dots & \\  \dots & \dots & \dots & \dots &   \end{array}  $                                                                              |                | $  \begin{array}{r rrrr}  & 2x^2 & \dots & \dots & \\  x & 2x^3 & \dots & \dots & \\  \dots & \dots & \dots & \dots &   \end{array}  $                                                                              |          |
| <p>•<sup>3</sup> is awarded for evidence of the cubic expression (which may be in the grid from part (a) ) AND the terms in the diagonal boxes summing to the second and third terms in the cubic respectively.</p> |                | <p>•<sup>3</sup> is awarded for evidence of the cubic expression (which may be in the grid from part (a) ) AND the terms in the diagonal boxes summing to the second and third terms in the cubic respectively.</p> |          |
| $  \begin{array}{r rrr}  & 2x^2 & 3x & -2 \\  x & 2x^3 & 3x^2 & -2x \\  +1 & 2x^2 & 3x & -2  \end{array}  $                                                                                                         |                | $  \begin{array}{r rrr}  & 2x^2 & x & -1 \\  x & 2x^3 & x^2 & -x \\  +2 & 4x^2 & 2x & -2  \end{array}  $                                                                                                            |          |
| $2x^2 + 3x - 2$                                                                                                                                                                                                     |                | $2x^2 + x - 1$                                                                                                                                                                                                      |          |
| $(x-1)(x+1)(2x-1)(x+2)$                                                                                                                                                                                             |                | $(x-1)(x+2)(x+1)(2x-1)$                                                                                                                                                                                             |          |
| Candidate E                                                                                                                                                                                                         |                | Candidate F                                                                                                                                                                                                         |          |
| $  \begin{array}{r rrrr}  \frac{1}{2} & 2 & 5 & 1 & -2 \\  & & 1 & 3 & 2 \\  \hline  & 2 & 6 & 4 & 0  \end{array}  $                                                                                                |                | $  \begin{array}{r rrrr}  \frac{1}{2} & 2 & 5 & 1 & -2 \\  & & 1 & 3 & 2 \\  \hline  & 2 & 6 & 4 & 0  \end{array}  $                                                                                                |          |
| $(x - \frac{1}{2})(2x^2 + 6x + 4)$                                                                                                                                                                                  |                | $(x - \frac{1}{2})(2x^2 + 6x + 4)$                                                                                                                                                                                  |          |
| $(2x-1)(x^2 + 3x + 2)$                                                                                                                                                                                              |                | $(x - \frac{1}{2})(2x+2)(x+2)$                                                                                                                                                                                      |          |
| $(x-1)(2x-1)(x+1)(x+2)$                                                                                                                                                                                             |                | $(x-1)(x - \frac{1}{2})(2x+2)(x+2)$                                                                                                                                                                                 |          |

| Question |     |  | Generic scheme                                                                                                                                                                                                                                                          | Illustrative scheme                                                                                                                                                                                                                                                                                                                                             | Max mark |
|----------|-----|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 11.      | (a) |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> use compound angle formula</li> <li>•<sup>2</sup> compare coefficients</li> <li>•<sup>3</sup> process for <math>k</math></li> <li>•<sup>4</sup> process for <math>a</math> and express in required form</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>k \cos x^\circ \cos a^\circ + k \sin x^\circ \sin a^\circ</math> stated explicitly</li> <li>•<sup>2</sup> <math>k \cos a^\circ = 1, k \sin a^\circ = \sqrt{3}</math> stated explicitly</li> <li>•<sup>3</sup> <math>k = 2</math></li> <li>•<sup>4</sup> <math>2 \cos(x - 60)^\circ</math></li> </ul> | 4        |

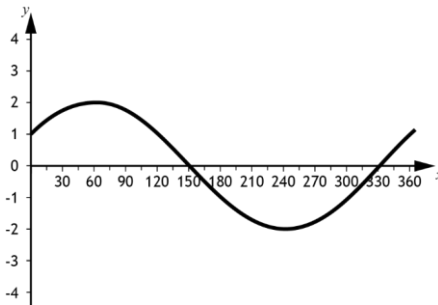
#### Notes:

- Accept  $k(\cos x^\circ \cos a^\circ + \sin x^\circ \sin a^\circ)$  for •<sup>1</sup>. Treat  $k \cos x^\circ \cos a^\circ + \sin x^\circ \sin a^\circ$  as bad form only if the equations at the •<sup>2</sup> stage both contain  $k$ .
- Do not penalise the omission of degree signs.
- $2 \cos x^\circ \cos a^\circ + 2 \sin x^\circ \sin a^\circ$  or  $2(\cos x^\circ \cos a^\circ + \sin x^\circ \sin a^\circ)$  is acceptable for •<sup>1</sup> and •<sup>3</sup>.
- <sup>2</sup> is not available for  $k \cos x^\circ = 1, k \sin x^\circ = \sqrt{3}$ , however •<sup>4</sup> may still be gained- see Candidate E
- <sup>3</sup> is only available for a single value of  $k, k > 0$ .
- <sup>3</sup> is not available to candidates who work with  $\sqrt{4}$  throughout parts (a) and (b) without explicitly simplifying at any stage. •<sup>4</sup> is still available.
- <sup>4</sup> is not available for a value of  $a$  given in radians.
- Candidates may use any form of the wave function for •<sup>1</sup>, •<sup>2</sup> and •<sup>3</sup>. However, •<sup>4</sup> is only available if the wave is interpreted in the form  $k \cos(x - a)^\circ$ .
- Evidence for •<sup>4</sup> may not appear until part (b).

#### Commonly Observed Responses:

| Candidate A                                                                                                                                                    | Candidate B - inconsistency                                                                                                                                                                                                  | Candidate C                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $k \cos x^\circ \cos a^\circ + k \sin x^\circ \sin a^\circ$ • <sup>1</sup>  | $k \cos x^\circ \cos a^\circ + k \sin x^\circ \sin a^\circ$ • <sup>1</sup>                                                                | $\cos x^\circ \cos a^\circ + \sin x^\circ \sin a^\circ$ • <sup>1</sup>  |
| $2 \cos a^\circ = 1$<br>$2 \sin a^\circ = \sqrt{3}$                                                                                                            | $\cos a^\circ = 1$<br>$\sin a^\circ = \sqrt{3}$                                                                                                                                                                              | $\cos a^\circ = 1$<br>$\sin a^\circ = \sqrt{3}$<br>$k = 2$                                                                                                   |
| $\tan a^\circ = \sqrt{3}$<br>$a = 60$                                                                                                                          | $\tan a^\circ = \sqrt{3}$<br>$a = 60$                                                                                                                                                                                        | $\tan a^\circ = \sqrt{3}$<br>$a = 60$                                                                                                                        |
| $2 \cos(x - 60)^\circ$ • <sup>4</sup>                                       | $2 \cos(x - 60)^\circ$ • <sup>3</sup>  • <sup>4</sup>  | $2 \cos(x - 60)^\circ$ • <sup>4</sup>                                   |

| Question                                                                                                                                                                                                                                                                                                                         | Generic scheme                                                                                                                                                                                                                                                                                                           | Illustrative scheme                                                                                                                                                                                                                    | Max mark |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>11. (a) (continued)</b>                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                        |          |
| <b>Candidate D - errors at ●<sup>2</sup></b><br>$k \cos x^\circ \cos a^\circ + k \sin x^\circ \sin a^\circ$ ● <sup>1</sup> ✓<br>$k \cos a^\circ = \sqrt{3}$<br>$k \sin a^\circ = 1$ ● <sup>2</sup> ✗<br>$\tan a^\circ = \frac{1}{\sqrt{3}}$<br>$a = 30$<br>$2 \cos(x - 30)^\circ$ ● <sup>3</sup> ✓ ● <sup>4</sup> ✓ <sub>1</sub> | <b>Candidate E - use of x at ●<sup>2</sup></b><br>$k \cos x^\circ \cos a^\circ + k \sin x^\circ \sin a^\circ$ ● <sup>1</sup> ✓<br>$k \cos x^\circ = 1$<br>$k \sin x^\circ = \sqrt{3}$ ● <sup>2</sup> ✗<br>$\tan x^\circ = \sqrt{3}$<br>$x = 60$<br>$2 \cos(x - 60)^\circ$ ● <sup>3</sup> ✓ ● <sup>4</sup> ✓ <sub>1</sub> | <b>Candidate F</b><br>$k \sin A \cos B + k \cos A \sin B$ ● <sup>1</sup> ✗<br>$k \cos A = 1$<br>$k \sin A = \sqrt{3}$ ● <sup>2</sup> ✗<br>$\tan A = \sqrt{3}$<br>$2 \cos(x - 60)^\circ$ ● <sup>3</sup> ✓ ● <sup>4</sup> ✓ <sub>1</sub> |          |

| Question |     |  | Generic scheme                                                                                                                                                                                                                                                                                       | Illustrative scheme                                                                                                                                                                                                                        | Max mark |
|----------|-----|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 11.      | (b) |  | <ul style="list-style-type: none"> <li>•<sup>5</sup> exactly two roots identifiable from graph</li> <li>•<sup>6</sup> coordinates of exactly two turning points identifiable from graph</li> <li>•<sup>7</sup> y-intercept and value of y at <math>x = 360</math> identifiable from graph</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>5</sup> (150,0) and (330,0)</li> <li>•<sup>6</sup> (60,2) and (240,-2)</li> <li>•<sup>7</sup> 1</li> </ul>  | 3        |

#### Notes:

10. •<sup>5</sup>, •<sup>6</sup> and •<sup>7</sup> are only available for attempting to draw a “cosine” graph with a period of  $360^\circ$ .
11. Ignore any part of a graph drawn outwith  $0 \leq x \leq 360$ .
12. Vertical marking is not applicable to •<sup>5</sup> and •<sup>6</sup>.
13. Candidate’s sketch in (b) must be consistent with the equation obtained in (a), see also Candidates G and H.
14. For any incorrect horizontal translation of the graph of the wave function arrived at in part (a) only •<sup>6</sup> is available.

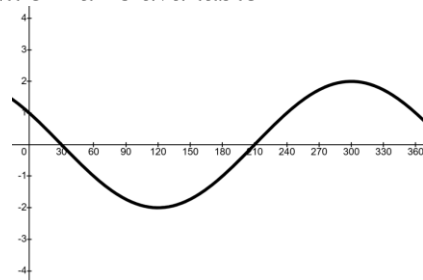
#### Commonly Observed Responses:

##### Candidate G - incorrect translation

- (a)  $2 \cos(x - 60)^\circ$  - correct equation
- (b) Incorrect translation:  
Sketch of  $2 \cos(x + 60)^\circ$   
only •<sup>6</sup> is available

##### Candidate H - incorrect equation

- (a)  $2 \cos(x + 60)^\circ$  - incorrect equation
- (b) Sketch of  $2 \cos(x + 60)^\circ$   
all 3 marks available



| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |  | Generic scheme                                                                                                                                                                                                                  | Illustrative scheme                                                                                                                                                                                                                                                                                                 | Max mark                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| 12.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  | <p>•<sup>1</sup> write in differentiable form</p> <p>•<sup>2</sup> differentiate</p> <p>•<sup>3</sup> solve for <math>a^{-\frac{2}{3}}</math> or <math>a^{\frac{2}{3}}</math></p> <p>•<sup>4</sup> solve for <math>a</math></p> | <p>•<sup>1</sup> <math>12x^{\frac{1}{3}}</math> stated or implied by •<sup>2</sup></p> <p>•<sup>2</sup> <math>12 \times \frac{1}{3} \times x^{-\frac{2}{3}}</math></p> <p>•<sup>3</sup> <math>a^{-\frac{2}{3}} = \frac{1}{4}</math> or <math>a^{\frac{2}{3}} = 4</math></p> <p>•<sup>4</sup> <math>a = 8</math></p> | 4                                                                                                                   |
| <b>Notes:</b> <ol style="list-style-type: none"> <li>•<sup>2</sup> is only available for differentiating a term with a fractional index.</li> <li>Where candidates attempt to integrate or make no attempt to differentiate, only •<sup>1</sup> is available.</li> <li>Accept <math>x^{-\frac{2}{3}} = \frac{1}{4}</math> or <math>x^{\frac{2}{3}} = 4</math> at •<sup>3</sup>. See Candidates A and B.</li> <li>•<sup>4</sup> is only available where the expression at •<sup>2</sup> is of the form <math>kx^{\frac{m}{n}}</math> where <math>m \neq 1</math>.</li> <li>Do not penalise the inclusion of <math>-8</math> at •<sup>4</sup>.</li> </ol> |  |  |                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                     |                                                                                                                     |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  |                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                     |                                                                                                                     |
| <b>Candidate A - working in terms of <math>x</math> throughout</b><br>...<br>$x^{-\frac{2}{3}} = \frac{1}{4}$<br>$x = 8$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |  | <p>•<sup>1</sup> ✓   •<sup>2</sup> ✓</p> <p>•<sup>3</sup> ✓</p> <p>•<sup>4</sup> ✗</p>                                                                                                                                          | <b>Candidate B</b><br>...<br>$x^{-\frac{2}{3}} = \frac{1}{4}$<br>$(x = 8)$<br>$a = 8$                                                                                                                                                                                                                               | <p>•<sup>1</sup> ✓   •<sup>2</sup> ✓</p> <p>•<sup>3</sup> ✓</p> <p>•<sup>4</sup> ✓</p>                              |
| <b>Candidate C</b><br>$f(x) = 12x^{\frac{3}{2}}$<br>$f'(x) = 18x^{\frac{1}{2}}$<br>$a^{\frac{1}{2}} = \frac{1}{18}$<br>$a = \frac{1}{324}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |  | <p>•<sup>1</sup> ✗</p> <p>•<sup>2</sup> ✓<sub>1</sub></p> <p>•<sup>3</sup> ✓<sub>1</sub></p> <p>•<sup>4</sup> ✓<sub>2</sub></p>                                                                                                 | <b>Candidate D - partly differentiated</b><br>$f(x) = 12x^{\frac{1}{3}}$<br>$f'(x) = 12 \times \frac{1}{3} x^{-\frac{2}{3}}$<br>$1 = 4a^{\frac{4}{3}}$<br>$\frac{1}{4} = a^{\frac{4}{3}}$<br>$a = \frac{1}{\sqrt[4]{8}}$                                                                                            | <p>•<sup>1</sup> ✓</p> <p>•<sup>2</sup> ✗</p> <p>•<sup>3</sup> ✓<sub>1</sub></p> <p>•<sup>4</sup> ✓<sub>2</sub></p> |



| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |  | Generic scheme                                                                                                                                                                                                                                   | Illustrative scheme                                                                                                                                                                                                          | Max mark |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 13.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | (a) |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> find midpoint of PQ</li> <li>•<sup>2</sup> find gradient of PQ</li> <li>•<sup>3</sup> find perpendicular gradient</li> <li>•<sup>4</sup> find equation of perpendicular bisector</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> (5,6)</li> <li>•<sup>2</sup> -4 or <math>-\frac{8}{2}</math></li> <li>•<sup>3</sup> <math>\frac{1}{4}</math></li> <li>•<sup>4</sup> <math>4y = x + 19</math></li> </ul> | 4        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |  |                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                              |          |
| 1. • <sup>4</sup> is only available as a consequence of using a perpendicular gradient and a mid-point.<br>2. The gradient of the perpendicular bisector must appear in fully simplified form at • <sup>3</sup> or • <sup>4</sup> stage for • <sup>3</sup> to be awarded.<br>3. At • <sup>4</sup> accept $4y - x = 19$ , $4y - x - 19 = 0$ , or any other rearrangement of the equation where the constant terms have been simplified.                                              |     |  |                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                              |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |  |                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                              |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | (b) |  | <ul style="list-style-type: none"> <li>•<sup>5</sup> identify x-coordinate of centre</li> <li>•<sup>6</sup> find y-coordinate of centre</li> <li>•<sup>7</sup> find radius</li> <li>•<sup>8</sup> state equation of circle</li> </ul>            | <ul style="list-style-type: none"> <li>•<sup>5</sup> 9</li> <li>•<sup>6</sup> 7</li> <li>•<sup>7</sup> <math>\sqrt{34}</math></li> <li>•<sup>8</sup> <math>(x-9)^2 + (y-7)^2 = 34</math></li> </ul>                          | 4        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |  |                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                              |          |
| 4. Do not accept "centre = (9,2)" as evidence of • <sup>5</sup> .<br>5. Where candidates use PQ, QR or PR as the diameter of the circle no marks are available.<br>6. • <sup>7</sup> and • <sup>8</sup> are only available as a consequence of using the point of intersection of two perpendicular bisectors and a point on the circumference of the circle.<br>7. Accept $r^2 = 34$ for • <sup>7</sup> .<br>8. $(x-9)^2 + (y-7)^2 = (\sqrt{34})^2$ does not gain • <sup>8</sup> . |     |  |                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                              |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |  |                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                              |          |
| <b>Candidate A - horizontal line through midpoint of PQ</b><br>Centre = (9,6)                      • <sup>5</sup> ✓   • <sup>6</sup> ✗<br>Radius = 5                            • <sup>7</sup> ✗<br>Equation: $(x-9)^2 + (y-6)^2 = 25$ • <sup>8</sup> ✗                                                                                                                                                                                                                             |     |  | <b>Candidate B - perpendicular bisector of PR</b><br>Perpendicular bisector of PR: $y = x - 2$<br>Centre = (9,7)                      • <sup>5</sup> ✓   • <sup>6</sup> ✓<br>⋮                                                                   |                                                                                                                                                                                                                              |          |

**[END OF MARKING INSTRUCTIONS]**